A Lightweight Trust Management Framework in the Networking Layer of IoT Devices

25_26J_009

Project Proposal Report

Bandara R.M.T

B.Sc. (Hons) in Information Technology Specializing in

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Acknowledgement

Working on this final-year research project has been an immensely enriching academic experience for me. Through the process of developing my research proposal, I have gained valuable insights into the research process, including how to formulate research questions, design suitable methodologies, and anticipate potential challenges.

I would like to express my heartfelt gratitude to my supervisor, Ms. Shashika Lokuliyana for her invaluable guidance, feedback, and support throughout this project. Her constant input and insightful suggestions are greatly appreciated. I would also like to express my gratitude to my co-supervisor, Mr. Anuradha Jayakody, for his unwavering guidance, feedback, and support throughout the project. I owe a debt of gratitude to all the lecturers at SLIIT as well.

I look forward to continuing to develop my research skills and contribute meaningful insights through this project.

Declaration

I declare that this is my own work, and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgment is made in the text.

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The above candidates are carrying out research for the undergraduate dissertation under my supervision.

Signature of the Supervisor: ..... Date 24/08/25

Signature of the Co-Supervisor: ..... Date 24/08/2025

Abstract

The Internet of Things (IoT) relies on interconnected, resource-constrained devices that exchange data over low-power networks. To enable efficient IPv6 communication in such environments, the IETF introduced 6LoWPAN (IPv6 over Low-Power Wireless Personal Area Networks). To address the challenges of these networks, the IETF ROLL working group standardized RPL (Routing Protocol for Low-Power and Lossy Networks) in RFC 6550[1], a lightweight IPv6-based routing protocol tailored for IoT. However, the protocol's simplicity makes it vulnerable to routing threats, particularly sinkhole[2] and selective forwarding[3] attacks, which undermine reliability and network performance. This research will propose a trust-aware routing mechanism that enhances RPL by incorporating trust and security metrics into routing decisions. Trust values will be derived from node behavior analysis, enabling dynamic and secure parent selection based on comprehensive trust evaluations. The approach aims to preserve RPL's lightweight design while strengthening trustworthiness and resilience against malicious routing behaviors. To achieve this, a trust computation framework, adaptive parent selection algorithms, and security mechanisms for detecting, mitigating, isolating, and recovering from sinkhole and selective forwarding attacks will be developed. The solution will be implemented and evaluated using the Contiki operating system and the Cooja network simulator, with the expectation of demonstrating significant improvements in performance and security for low-power and lossy IoT environments.

Key word: RPL, Lightweight trust-Aware Routing, Sink hole attack, selective forward attack,

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1.0 Introduction

1.1 Background

The Internet of Things (IoT) continues to revolutionize modern computing by interconnecting diverse, resource-constrained devices across domains like smart cities, healthcare, home automation, and industrial monitoring. This vision builds on the foundation of ubiquitous computing first conceptualized by Weiser (1991) [1]and has matured through key innovations in wireless sensor networks and IPv6 integration. The evolution from isolated embedded systems to interconnected IoT ecosystems has been driven by advances in wireless sensor networks (WSNs), low-power microcontrollers, and standardized communication protocols, enabling the deployment of billions of connected devices with severe constraints on processing power, memory, and energy resources.

To address the fundamental challenge of IPv6 communication over low-power wireless networks, the Internet Engineering Task Force (IETF) standardized IPv6 over Low-Power Wireless Personal Area Networks (6LoWPAN) in RFC 4944 [2]This adaptation layer enables IPv6 packets to be transmitted over IEEE 802.15.4 networks through header compression techniques that reduce the standard 40-byte IPv6 header to as little as 2 bytes, fragmentation and reassembly mechanisms for handling large packets, and neighbor discovery optimizations for constrained environments. Building upon this foundation, the IETF's Routing Over Low-Power and Lossy Networks (ROLL) working group recognized that traditional routing protocols like OSPF, RIP, and AODV were fundamentally unsuitable for Low-Power and Lossy Networks (LLNs) due to their high control overhead, excessive memory requirements, and poor convergence characteristics in lossy environments.

Consequently, the ROLL working group developed and standardized the IPv6 Routing Protocol for Low-Power and Lossy Networks (RPL) in RFC 6550 [3], providing a specialized routing solution tailored for the unique constraints of IoT devices. RPL creates Destination-Oriented Directed Acyclic Graph (DODAG) topologies optimized for the many-to-one traffic patterns common in IoT applications, where multiple sensor nodes report data to central collection points. The protocol employs a sophisticated control message framework using ICMPv6 packets, including DODAG Information Object (DIO) messages for topology discovery, Destination Advertisement Object (DAO) messages for route establishment, and DODAG

Information Solicitation (DIS) messages for information requests. RPL's flexibility is further enhanced through Objective Functions (OFs) such as OF0 [4] for basic connectivity and MRHOF [5] for metric-based optimization, allowing customization for different application requirements while maintaining the protocol's lightweight characteristics.

However, RPL's design prioritized energy efficiency and simplicity over security, making it vulnerable to various routing attacks [9], [11],[13] that exploit the protocol's trust assumptions and cooperative nature. The lightweight design that enables operation on resource-constrained devices also creates security vulnerabilities, particularly against insider attacks from compromised nodes that can manipulate routing information, selectively forward packets, or create false routing advertisements. As IoT deployments have scaled from research environments to critical infrastructure applications in smart grids, healthcare monitoring, and industrial control systems, the security limitations of RPL have become increasingly apparent, necessitating the development of trust-aware routing mechanisms 7], [10], [12], [15] that can detect and mitigate malicious behavior while preserving the protocol's fundamental operational characteristics.

1.2 Literature Survey

Trust-Based Security Approaches

The integration of trust mechanisms into RPL routing has emerged as prominent research direction for addressing security vulnerabilities while maintaining lightweight operation. The SMTrust protocol [10], [12] represents a comprehensive hybrid trust framework that evaluates nodes based on mobility patterns, link stability, energy consumption, historical behavior, and neighbor recommendations. This multi-dimensional approach demonstrates significant improvements against rank and blackhole attacks, achieving 46% higher topology stability, 45% lower packet loss, and 35% higher throughput compared to standard RPL, while introducing only 2.3% energy overhead. However, the computational complexity of SMTrust's trust aggregation mechanisms may limit its scalability in large networks, and its evaluation focuses primarily on individual attack types rather than combined attack scenarios.

Specialized defenses against selective forwarding attacks have been developed by Arshad et al. [13], who propose an advanced trust-based detection model that achieves 96% detection accuracy with minimal energy overhead (~3.4%) even under dynamic and stealthy attack conditions. Their

approach employs statistical analysis of packet forwarding behavior and neighbor monitoring to identify malicious nodes, demonstrating effectiveness against sophisticated selective forwarding strategies. Nevertheless, this solution lacks comprehensive evaluation against combined attack vectors where adversaries simultaneously employ multiple attack strategies to evade detection mechanisms.

Machine Learning and Adaptive Approaches

The application of artificial intelligence techniques to IoT routing security has gained momentum with the development of adaptive learning mechanisms. The iTRPL protocol [7] integrates multi-agent reinforcement learning (MARL) with dynamic trust updates, enabling nodes to cooperatively learn optimal routing decisions while rewarding or penalizing neighbors based on observed behavior. This approach demonstrates self-adaptive, intelligent protection against insider threats by continuously updating trust models based on network interactions. While iTRPL shows promising results in adaptive threat response, the computational and memory requirements of maintaining reinforcement learning models may pose challenges for deployment on severely resource-constrained IoT devices with limited processing capabilities and memory (typically 8-32KB RAM).

Alternative Security Frameworks

Parallel research efforts have explored alternative security enhancement approaches beyond traditional trust mechanisms. The Chained Secure Mode (CSM) framework [8] integrates network coding techniques with RPL secure modes to effectively counter wormhole, neighbor spoofing, and CloneID attacks. CSM demonstrates improved packet delivery ratios and reduced latency and energy consumption compared to existing secure modes, offering a complementary approach to trust-based solutions. However, CSM requires modifications to the standard RPL protocol stack and may introduce compatibility challenges with existing IoT deployments.

Lightweight Detection Mechanisms

Recent research has emphasized the development of computationally efficient security solutions suitable for resource-constrained environments. Al-Sarawi *et al.* [9] proposed a rule-based passive sinkhole detection mechanism that minimizes computational complexity while preserving detection accuracy through pattern analysis of routing behavior. Similarly, Adarbah *et al.* [13] developed parent attack mitigation strategies that focus on lightweight monitoring and response mechanisms.

While these approaches successfully reduce computational overhead, they typically address specific attack types and may lack comprehensive protection against sophisticated combined attack scenarios.

Broader Security Considerations

Extended trust frameworks have been developed to address diverse threat models beyond traditional routing attacks. Muzammal *et al.* [12] proposed trust-based secure routing frameworks that extend trust metrics beyond conventional routing misbehaviors to include denial-of-service attack patterns and resource exhaustion attacks. Similarly, Hashemi and Aliee [15] introduced a fuzzy, dynamic, and trust-based routing protocol that enhances resilience against a wide range of adversarial behaviors. This broader perspective on trust evaluation provides more comprehensive threat coverage but introduces additional complexity in trust computation and may require more sophisticated monitoring mechanisms.

Critical Analysis and Limitations

While the surveyed literature demonstrates significant advances in RPL security enhancement, several critical limitations persist across existing approaches. Most trust-based solutions focus on individual attack types rather than realistic combined attack scenarios, creating potential vulnerabilities against sophisticated adversaries. Computational scalability remains a concern for complex trust frameworks, particularly in large-scale deployments where trust computation overhead may exceed the capabilities of resource-constrained devices. Additionally, real-time adaptability is limited in many existing solutions, which typically employ fixed update intervals or reactive mechanisms that may introduce significant delays in threat detection and response. The lack of standardized evaluation frameworks further complicates the assessment and comparison of different approaches, making it challenging to determine optimal solutions for specific deployment scenarios.

1.3 Research Gap

While existing research demonstrates promising advances in trust-aware routing for IoT environments, several critical gaps persist that limit the practical deployment and effectiveness of current solutions. Computational scalability represents a fundamental limitation across most existing trust frameworks. For instance, while SMTrust [10] achieves impressive security metrics with only 2.3% energy overhead in small-scale scenarios, the computational complexity of its multi-

dimensional trust aggregation increases quadratically with network size, making it unsuitable for large-scale deployments exceeding 200 nodes.

Similarly, the iTrust protocol [7], despite its adaptive capabilities, requires significant memory and processing resources for maintaining learning models across multiple agents, creating deployment barriers for resource-constrained IoT devices with limited RAM (typically 8–32 KB) and processing capabilities.

Attack resilience limitations constitute another significant gap in current research. Most existing solutions focus on individual attack types rather than sophisticated combined attack scenarios that represent realistic threat models in adversarial environments. The advanced trust-based detection model proposed in [8] demonstrates excellent performance against selective forwarding attacks with 96% detection accuracy but lacks evaluation against combined sinkhole–selective forwarding attacks where malicious nodes first attract traffic through false routing advertisements and then selectively drop packets to maximize damage while evading detection.

Similarly, the CSM framework [11] effectively counters wormholes and spoofing attacks but does not address scenarios where attackers dynamically switch between different attack strategies to circumvent trust-based detection mechanisms. This limitation is particularly concerning given that sophisticated attackers in real-world scenarios typically employ multiple attack vectors simultaneously to maximize network disruption.

Real-time adaptability deficiencies represent another critical gap where existing trust mechanisms fail to provide dynamic response to evolving attack patterns and network conditions. Current solutions typically employ fixed trust update intervals or reactive threshold-based mechanisms that introduce significant delays in threat detection and mitigation. For example, while the DoS-aware trust framework [12] extends trust metrics beyond traditional routing misbehaviors, its trust recalculation occurs at predetermined intervals (typically every 30–60 seconds), creating vulnerability windows where sophisticated attackers can exploit the system before trust scores are updated.

Furthermore, most existing approaches lack adaptive trust decay mechanisms that can dynamically adjust trust degradation rates based on attack intensity and network criticality, resulting in either overly conservative trust management that impedes legitimate network operations or aggressive trust updates that create instability under normal network variations.

Implementation and evaluation gaps further compound these limitations, with most research focusing on theoretical frameworks without addressing practical deployment challenges in real-world IoT environments. Existing solutions often require significant modifications to standard RPL protocol implementations, creating compatibility issues with legacy infrastructure and standardized IoT platforms. Additionally, current evaluation methodologies typically assess security and performance in isolation rather than providing comprehensive analysis of the trade-offs between security enhancement, energy consumption, memory utilization, and network performance under diverse operational conditions. The lack of standardized evaluation frameworks makes it difficult to compare different approaches objectively and assess their suitability for specific deployment scenarios.

Integration and interoperability challenges represent an often-overlooked gap where existing trust-aware solutions fail to consider compatibility with standard RPL objective functions (OF0 and MRHOF) and integration with other IoT security mechanisms. Most current research proposes standalone trust frameworks without addressing how these solutions can be seamlessly integrated into existing IoT deployments or how they interact with other security measures such as link-layer encryption or application-layer authentication. This lack of holistic security consideration limits the practical adoption of trust-based routing solutions in production IoT environments where multiple security layers must coexist effectively.

These identified gaps collectively highlight the need for a comprehensive trust-aware routing solution that addresses scalability, multi-attack resilience, real-time adaptability, and practical deployment considerations while maintaining compatibility with standard RPL implementations and existing IoT infrastructure.

1.4 Research Problem

Despite significant advances in securing RPL through trust-based mechanisms, existing solutions suffer from critical limitations that hinder their practical deployment in real-world IoT environments. Current trust-aware routing approaches face three fundamental challenges: computational overhead scalability, where trust computation complexity increases exponentially with network size, making them unsuitable for large-scale deployments [10], [7]; limited attack resilience, as most solutions focus on individual attack types (either sinkhole or selective forwarding) rather than sophisticated combined attack scenarios that represent realistic threat models [8], [11]; and

inadequate real-time adaptability, where trust updates occur at fixed intervals or reactive thresholds, failing to provide dynamic response to evolving attack patterns and network conditions [12].

Furthermore, existing trust frameworks often require significant modifications to the standard RPL protocol stack, creating deployment barriers and compatibility issues with legacy IoT infrastructure. The lack of comprehensive evaluation frameworks that consider both security effectiveness and resource consumption under diverse attack scenarios compounds these challenges, making it difficult to assess the practical viability of proposed solutions. This research addresses the critical need for a lightweight, dynamic, and comprehensive trust-aware routing mechanism that maintains RPL's fundamental design philosophy while providing robust protection against combined routing attacks in resource-constrained IoT environments.

In addition, current solutions lack comprehensive frameworks for managing the post-detection phase of security incidents. Existing research primarily focuses on attack identification but fails to address critical challenges in coordinated node isolation, attack impact mitigation, and network recovery. Limitations include the absence of distributed consensus mechanisms for isolation decisions, insufficient strategies for maintaining network connectivity during malicious node removal, and inadequate recovery protocols capable of restoring optimal network topology without prolonged service disruption. This research therefore addresses the need for a holistic post-detection security management framework that ensures effective isolation of confirmed malicious nodes, minimizes attack impact during response procedures, and enables rapid network recovery while preventing re-infiltration by previously isolated attackers.

1.5 Research Questions

This research is guided by the following specific research questions:

Research Questions 1: How can malicious nodes be effectively isolated from the network once identified by the trust-based detection system while maintaining network connectivity and preventing false isolation of legitimate nodes?

Research Questions 2: How can mitigation strategies be implemented to minimize the impact of ongoing attacks and prevent attack propagation while the isolation process is being executed?

Research Questions 3: How can the network topology be efficiently reconstructed and normal operations restored after malicious nodes have been isolated, ensuring minimal disruption to critical IoT services?

Research Questions 4: What also mechanisms are needed to prevent isolated malicious nodes from rejoining the network and launching subsequent attacks through identity spoofing or trust rehabilitation exploitation?

Research Questions 5: How can the isolation, mitigation, and recovery processes be optimized to operate within the resource constraints of IoT devices while maintaining effectiveness against sophisticated attack scenarios?

2.0 Objective

2.1 Main Objective

The main objective of this research is to design, develop, and evaluate a lightweight, trust-aware routing mechanism for the IPv6 Routing Protocol for Low-Power and Lossy Networks (RPL) that provides comprehensive protection against sinkhole and selective forwarding attacks in IoT environments. The proposed solution integrates multi-dimensional trust computation, dynamic parent selection, accurate attack identification, effective node isolation, mitigation, and recovery strategies. By balancing security with energy efficiency, scalability, and compatibility with standard RPL implementations, the research seeks to achieve measurable improvements in attack detection accuracy, network reliability, and resilience under adversarial conditions. The final outcome will be validated through Contiki/Cooja simulations and experimental testbeds, demonstrating practical feasibility for real-world IoT applications such as smart cities, healthcare monitoring, industrial automation, and critical infrastructure, while contributing both theoretical advancements and deployable security solutions.

2.2 Sub objectives

Sub objective 1: Design Distributed Malicious Node Isolation Framework

To develop a distributed isolation protocol that enables coordinated removal of identified malicious nodes from active routing participation. This framework must implement secure consensus mechanisms for isolation decisions, maintain blacklist consistency across network nodes, prevent legitimate nodes from being falsely isolated due to temporary network issues or incomplete information, and ensure isolation decisions can be executed within 2-3 seconds after attack confirmation from the detection system.

Sub objective 2: Implement Attack Impact Mitigation Strategies

To create immediate mitigation mechanisms that limit attack damage while isolation is being processed. This includes developing traffic redirection protocols to route packets away from suspected nodes, implementing temporary quarantine states for nodes under investigation, creating emergency communication channels that bypass compromised routing paths, and ensuring critical IoT services remain operational during attack response with minimal performance degradation (<10% throughput reduction).

Sub objective 3: Develop Network Recovery and Topology Reconstruction Algorithms

To design comprehensive recovery mechanisms that restore optimal network functionality after malicious node isolation. This includes implementing rapid alternative route discovery for network segments affected by node removal, developing topology healing algorithms that reconnect isolated network partitions, creating load balancing mechanisms to redistribute traffic after topology changes, and ensuring recovery processes complete within 30 seconds to minimize service disruption.

Sub objectives 4: Establish Secure Re-authentication and Trust Rehabilitation Framework

To implement robust mechanisms that prevent isolated malicious nodes from rejoining the network through identity spoofing or exploitation of trust rehabilitation processes. This includes developing multi-factor node authentication protocols, implementing progressive trust rebuilding processes for nodes seeking network re-entry, creating secure node identity verification mechanisms, and establishing monitoring periods for rehabilitated nodes with enhanced scrutiny.

Sub objectives 5: Validate Integration with Detection and Trust Systems

To demonstrate seamless integration of isolation, mitigation, and recovery mechanisms with colleagues' attack detection and trust computation systems. This includes implementing standardized interfaces for receiving attack notifications, validating coordinated response with dynamic parent selection mechanisms, ensuring compatibility with trust metric updates during recovery, and conducting comprehensive testing of the complete integrated system using Contiki OS and Cooja simulator

3.0 Methodology

This research employs a structured five-phase methodology to design, implement, and validate mechanisms for isolating compromised nodes, mitigating the impact of routing attacks, and ensuring secure recovery in trust-aware RPL routing. Each phase produces specific deliverables and builds towards an integrated, validated framework that operates seamlessly with detection, trust, and parent selection modules developed by collaborators.

3.1 Research Approach / Strategy

This research adopts a simulation-based experimental approach to investigate and validate the proposed lightweight trust–security integrated framework for RPL-based IoT networks. Physical IoT testbeds, while valuable, are costly, resource-intensive, and limited in scalability, making them unsuitable for iterative experimentation within the scope of this study. In contrast, simulation environments such as Contiki-NG with the Cooja simulator provide a reproducible, flexible, and cost effective alternative that allows controlled experimentation across networks ranging from 20 to 100 nodes.

The simulation approach enables systematic evaluation under both normal operating conditions and adversarial scenarios (sinkhole and selective forwarding attacks). The performance of the framework will be assessed using a structured Evaluation Matrix that defines criteria across four dimensions: Detection Effectiveness, Network Performance, Resource Efficiency, and System Robustness & Integration. Specific targets include achieving at least 95% detection accuracy, ≤5% false positives, ≥90% packet delivery ratio under attack, ≥98% post-recovery PDR, ≤10% additional energy consumption, ≤15% CPU/RAM overhead, and a ≤5s recovery time for stable DODAG reconstruction. The framework will also be evaluated for scalability across networks of

20, 50, and 100 nodes, and for module interoperability to ensure seamless coordination between trust, detection, and recovery components.

This strategy ensures that the study remains lightweight and feasible for constrained IoT devices, while still providing robust, reproducible evidence of the framework's practicality. By combining theoretical modeling with repeatable simulation experiments, the research delivers a credible blueprint that other researchers can replicate and extend in future IoT security studies.

3.2 System diagram

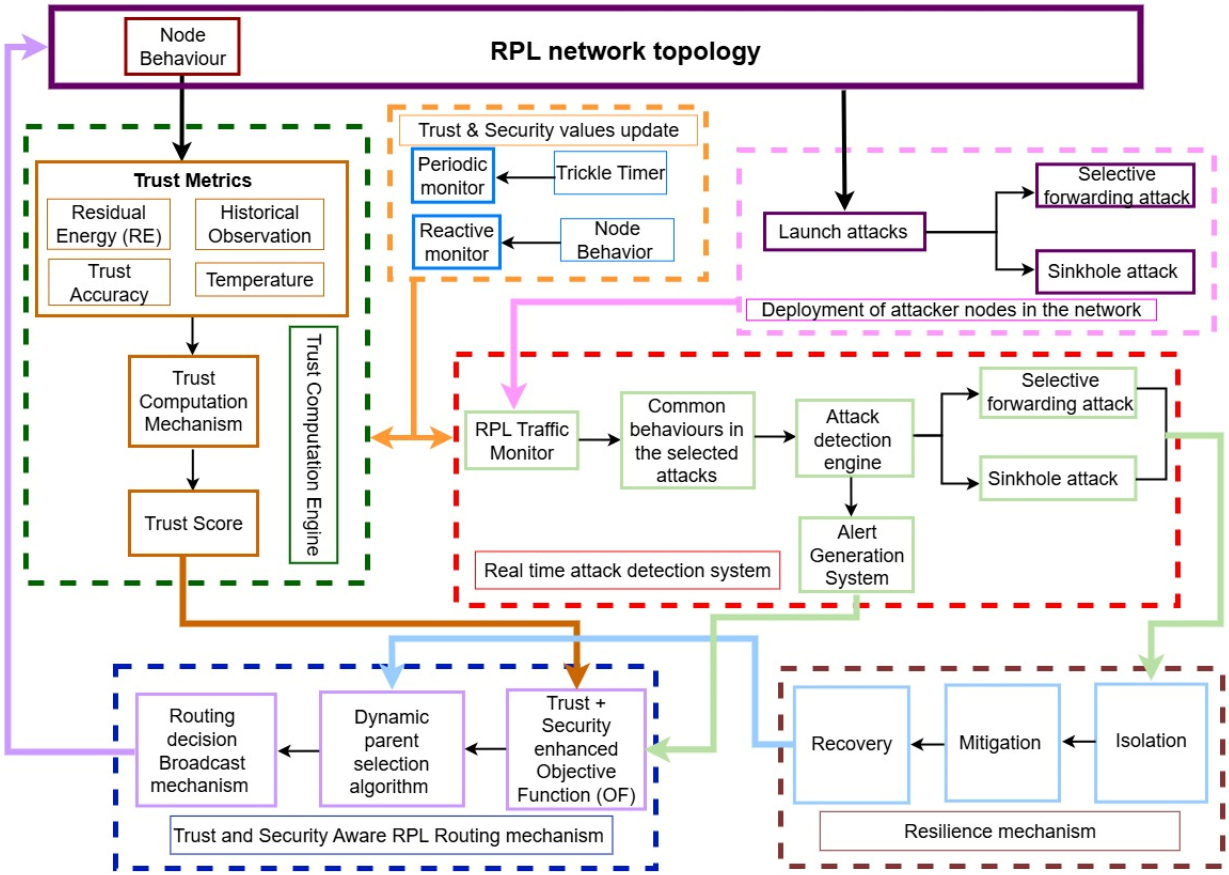

Fig 1: Lightweight Trust Management Framework for RPL-based IoT Networks

The proposed system follows a multi-layered architecture that integrates trust computation, real-time attack detection, secure routing, and resilience mechanisms within an RPL-based IoT topology. The high-level architecture is illustrated in the above figure.

RPL Network Topology

The foundation of the system is the RPL DODAG (Destination Oriented Directed Acyclic Graph) structure. From this network layer, both control and data messages are collected for monitoring. These include DIO messages (Rank, ETX, DODAG version, preferred parent), DIS messages (solicitation frequency and timing patterns), DAO messages (route updates), and data packets with headers (source, destination, sequence number, timestamps). Additionally, network metrics such as RSSI, ETX, success rate, energy level, and link stability are captured. This provides a comprehensive traffic profile necessary for attack detection [16], [18].

Trust Computation Engine

The system computes trust scores for nodes based on multiple lightweight metrics: Residual Energy, Temperature, Historical Observation, and Trust Accuracy. Trust scores are updated by two complementary monitoring schemes:

a. Periodic Monitor: Operates on trickle timers to evaluate nodes at regular intervals.
b. Reactive Monitor: Triggered when anomalous or suspicious behavior is observed.

This dual-monitoring approach ensures both proactive and reactive adaptation to dynamic IoT environments [19], [22].

Attack Simulation

To evaluate robustness, sinkhole and selective forwarding attacks are launched in a simulated RPL network. In sinkhole attacks, a malicious node falsifies rank or ETX values to attract traffic, while in selective forwarding, malicious nodes drop packets selectively while forwarding others to avoid detection [23], [25]. These attacks are implemented in Contiki OS / Cooja, allowing traffic trace collection under both normal and adversarial conditions.

RPL Traffic Monitor and Feature Extraction

Traffic monitoring enables the extraction of common behavioral patterns between sinkhole and selective forwarding attacks. Potential lightweight features include: CDFR (Cumulative Drop Forwarding Ratio), PTI (Packet Timing Irregularity), SIS (Sequence Integrity Score), RMC (Route Metric Coherence), and PCRE (Parent-Change Ripple Effect). Since the focus is on lightweight

operation, only a subset of these features will be selected based on relevance and efficiency [26], [27].

- CDFR: How many controls vs. data packets each node sends
- PTI: Are packets arriving at weird times?
- SIS: Are sequence numbers missing/jumbled?
- RMC: Are advertised network metrics consistent?
- PCRE: Are many nodes suddenly changing parents?

Attack Detection Engine

The detection engine applies statistical thresholding with confidence scoring, avoiding complex ML techniques to maintain lightweight operation. By comparing extracted features against thresholds, the system identifies anomalies in real time, classifies the attack type (sinkhole vs. selective forwarding), and pinpoints suspicious nodes.

Sinkhole Attack in RPL:

In a sinkhole attack, a malicious node advertises falsified routing metrics such as low rank, high link quality, or attractive paths to lure neighboring nodes into routing their traffic through it. Once established as a preferred parent, the attacker can drop, delay, or manipulate packets, thereby disrupting the RPL topology and misleading the DODAG structure [16], [19], [21], [33].

Selective Forwarding Attack in RPL:

In a selective forwarding attack, a compromised node that is already on a valid route deliberately drops a subset of the packets while forwarding others. Unlike a blackhole attack, the malicious node forwards enough traffic to maintain trust and avoid immediate detection but strategically discards critical or time-sensitive packets to degrade network performance and reliability [25], [30], [35], [36].

Alert Generation System

Upon detection, alerts are generated with details of the attack type and malicious node ID. Alerts are forwarded to:

a. The Trust & Security-Aware Routing module, for rerouting decisions.

b. The Resilience mechanism, for isolation, mitigation, and recovery.

Trust & Security-Aware Routing

Routing decisions are enhanced by combining trust scores with attack detection alerts. The module supports:

a. Dynamic parent selection that avoids malicious nodes.
b. Routing broadcast updates to reconfigure safe paths.
c. Enhanced Objective Function (OF) that integrates trust and security metrics into RPL route optimization [30].

Resilience Mechanism

The resilience layer ensures network stability after attack detection. It includes:

Isolation: Quarantining attacker nodes.

Mitigation: Updating routing tables to bypass malicious links.

Recovery: Triggering RPL healing and DODAG restoration to maintain QoS.

3.3 Work Breakdown Structure and Time line

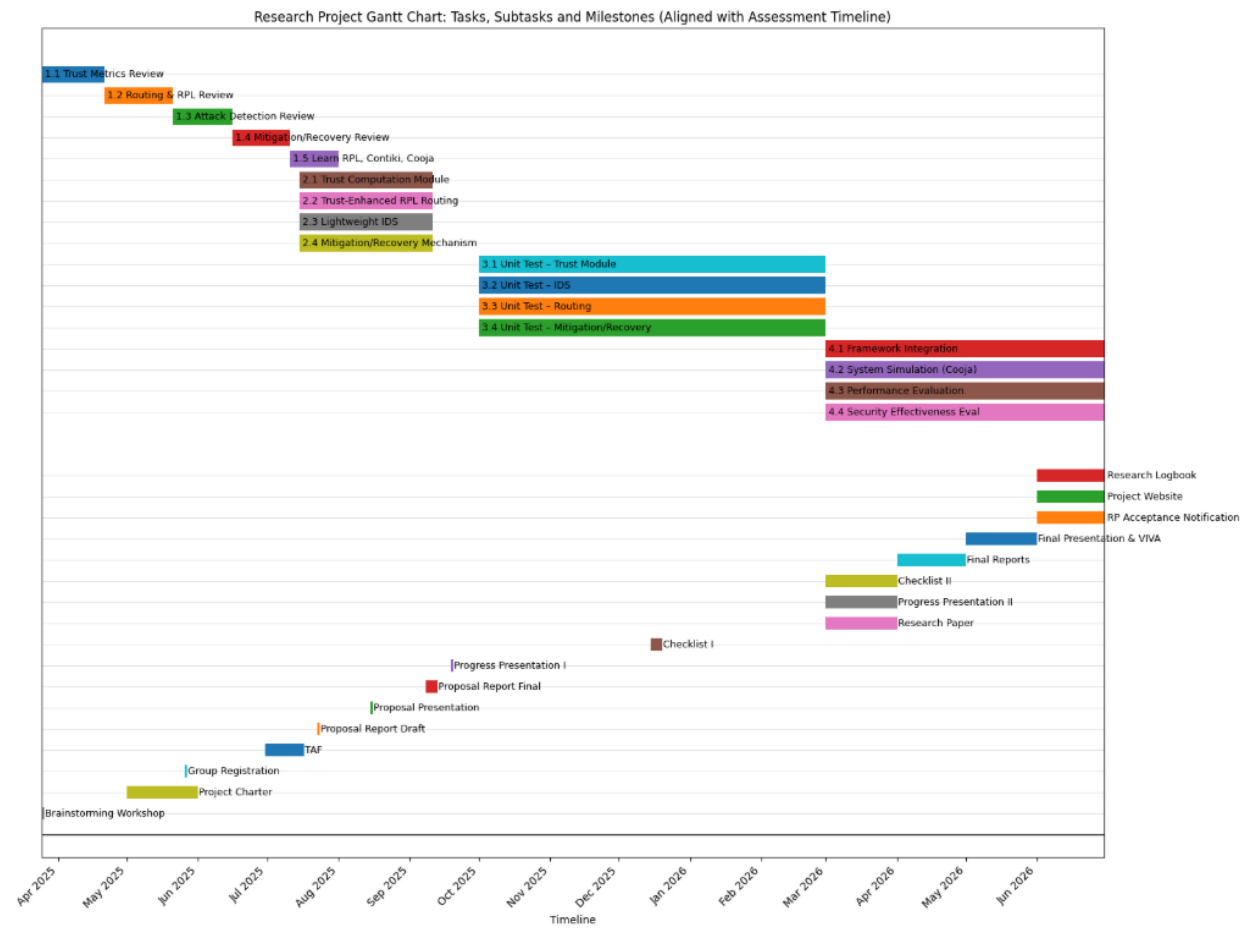

Fig2; Work Breakdown Structure and Time line

The Gantt chart presents the complete work breakdown structure (WBS) and timeline for the proposed research. It maps major tasks and subtasks to the five specific objectives, showing their sequencing and overlap. Key academic milestones including proposal submission, progress presentations, research paper, final report, and viva are aligned with the tentative assessment schedule. This visualization highlights dependencies, workload distribution across members, and the integrated flow of research activities.

Task 1: Research Foundation

- 1.1 Literature Review
- 1.2 Gap Analysis & Problem Definition

- 1.3 Proposal Writing & Submission

Task 2: Component Development *(aligned with specific objectives)*

- 2.1 Trust Computation Engine
- 2.2 Trust-Enhanced RPL Routing
- 2.3 Intrusion Detection System (IDS)
- 2.4 Isolation, Mitigation & Recovery Mechanism

Task 3: Framework Integration

- 3.1 Module Integration (Trust + IDS + Routing + Recovery)
- 3.2 System Debugging & Refinement

Task 4: Evaluation & Validation

- 4.1 Simulation Setup (Contiki OS / Cooja)
- 4.2 Performance Testing (PDR, Latency, Energy, Accuracy)
- 4.3 Security Effectiveness Testing (Sinkhole + Selective Forwarding)
- 4.4 Benchmarking vs. State-of-the-Art

4.0Project Requirements

4.1 Functional Requirements

a. Ability to configure an IoT network using Cooja/Contiki with the RPL protocol to ensure the environment is set up accurately for simulating IoT scenarios with RPL, forming the basis for testing security mechanisms and attacks.

b. Ability to simulate sinkhole and selective forwarding attacks on the RPL network. The system must realistically replicate these specific routing attacks, allowing for the evaluation of their impact and the effectiveness of defense strategies.

c. Ability to isolate the malicious nodes (detected by teammate nodes) that are involved in sinkhole and selective forwarding attacks, using trust-aware security mechanisms to prevent them from affecting the network.

d. Ability to mitigate ongoing threats in real time by dynamically rerouting traffic through trusted paths, preventing isolated or compromised nodes from disrupting communication, and ensuring continuous network operation.

e. Ability to restore network integrity after an attack by re-establishing secure and trusted routes, maintaining service availability, and ensuring overall system resilience against future threats.

f. Ability to monitor trust scores, routing paths, and network performance metrics relevant to attack detection and network health, with continuous observation of trustworthiness and routing behavior enabling early detection of anomalies and assessment of the network's security posture.

g. Ability to compare baseline RPL routing performance with the proposed secure, trust-aware RPL implementation, ensuring that performance and security evaluations are measurable against standard RPL to demonstrate the added value of the proposed solutions.

4.2 User Requirements

The proposed trust-aware RPL framework must satisfy the expectations of IoT network administrators, system operators, and security teams. From their perspective, the essential requirements are as follows:

A. Stable Connectivity and Recovery

- Users expect the IoT network to maintain stable connectivity even in the presence of routing attacks. When attacks such as sinkhole or selective forwarding occur, the system must ensure that network services remain available without long-term disruption. To achieve this, the framework should implement secure recovery mechanisms that allow legitimate nodes to rejoin the routing topology after malicious nodes are isolated.

This ensures that users experience continuous and reliable communication despite adversarial conditions.

B. Energy efficiency

- Given that IoT devices are inherently resource-constrained, energy efficiency is a critical user requirement. Users expect that the introduction of security mechanisms will not lead to excessive energy consumption or resource depletion. To meet this expectation, the proposed framework must remain lightweight in design, minimizing computational and communication overhead while still providing robust protection. This balance ensures that the network remains sustainable and practical for real-world IoT deployments.

C. Accuracy and Reliability

- The most significant requirement for users is that the system delivers accurate routing decisions and reliable data transmission. Inaccuracies caused by compromised nodes can result in data loss, corruption, or misrouting, which are unacceptable in critical IoT applications. The framework must therefore employ trust-based mechanisms to evaluate node behavior, ensuring that only trustworthy nodes are selected for routing. By doing so, packet delivery ratio and data integrity are preserved, guaranteeing reliability under both normal and adversarial conditions.

D. Attack Protection Mechanisms

- Users require effective protection against common RPL vulnerabilities, particularly sinkhole and selective forwarding attacks. The proposed framework must therefore include defensive mechanisms capable of detecting malicious routing behavior, isolating the compromised nodes, and mitigating their negative effects on the network. By preventing attackers from exploiting the routing topology, the system ensures secure communication paths and protects the overall health of the IoT network.

E. Trusted RPL Network

- Ultimately, users demand a trusted IoT network where routing decisions are based on the trustworthiness of participating nodes. This requires the system to dynamically select communication paths that favor trusted parent nodes while excluding malicious or suspicious ones. Through this trust-based parent selection process, the framework

guarantees that IoT communication remains secure, resilient, and dependable. Such a trusted RPL network aligns with user expectations for accuracy, security, and stability in mission-critical applications.

4.3System Requirements

A. The system shall calculate trust scores for IoT nodes based on behavioral metrics, providing a quantitative measure of node reliability and forming the foundation of trust-aware routing decisions.

B. The framework is required to dynamically adjust parent selection according to trust values, ensuring that compromised nodes are excluded from routing paths and overall network resilience is improved.

C. It must be capable of detecting sinkhole and selective forwarding attacks with high accuracy, thereby minimizing false positives and negatives while ensuring that genuine nodes are not misclassified as malicious.

D. The system shall isolate malicious nodes and mitigate their impact on routing, preventing further disruption and maintaining the integrity and stability of the network.

E. It is necessary to ensure secure recovery so that normal nodes can rejoin the network after mitigation, as recovery mechanisms are critical for restoring operation and sustaining connectivity.

F. The system should log results, including trust values, routing changes, and performance metrics, to provide traceability and enable researchers or administrators to analyze performance, validate effectiveness, and refine defense mechanisms.

4.4Non-Functional Requirements

a. Scalability must handle networks of 50–100 nodes in simulation without severe performance degradation.

b. Lightweight design must minimize additional computation and communication overhead (<10% compared to standard RPL).

c. Accuracy, attack detection accuracy ≥ 90% with low false positive rate.

d. Reliability, Ensure ≥ 95% packet delivery ratio under attack scenarios.

e. Resilience, Support recovery from node isolation without disrupting overall network connectivity.

f. Energy efficiency must preserve energy efficiency of constrained IoT devices.

g. Standards compatibility must remain compatible with RPL as defined in RFC 6550.

4.5 Software Requirements

Requirement	Software
A. Primary Operating system	Ubuntu latest version
B. IOT Operating System	Contiki-NG OS.
C. IOT Simulator	Cooja Network Simulator.
D. Packet Analysis	Wireshark
E. Scripting/Analysis	Python
F. Documentation	MS Word
G. Version controlling	GitHub
H. Communication	Slack
I. Project management	Jira

4.6Hardware Requirements

Since this project is based entirely on a simulation environment, sufficient computational power is essential to implement an IoT network with a considerable number of nodes (50–100) and to simulate security threats such as sinkhole and selective forwarding attacks. Therefore, the hardware platform must be capable of handling resource-intensive simulations while ensuring stability and performance. To meet these needs, high-performance

simulation environment is required. Environment should run on a laptop with the following minimum specifications:

a. Processor: Intel Core i7 (or equivalent)
b. Memory: 16 GB RAM
c. Storage: 512 GB SSD

4.7 Expected Test Cases

A. Baseline Test

- Evaluate the normal operation of the RPL protocol in a secure environment without any malicious activity. This establishes a performance benchmark for metrics such as packet delivery ratio, latency, and control overhead.

B. Sinkhole Attack Test

- Assess the effect of a sinkhole attack, where a compromised node advertises falsified routing information to attract traffic, allowing observation of its impact on data forwarding and network reliability.

C. Selective Forwarding Attack Test

- Measure the network's performance when malicious nodes intentionally drop selected packets, enabling analysis of data delivery degradation and routing instability under this adversarial condition.

D. Mitigation Test

- Validate the effectiveness of the proposed trust-aware detection and isolation mechanism. The system is tested under attack scenarios to confirm its ability to identify malicious nodes, isolate them, and maintain routing efficiency.

E. Recovery Test

- Examine how the network restores stability after mitigation, focusing on the reintegration of legitimate nodes, reconstruction of the DODAG, and improvement in performance metrics following attack isolation.

4.8 Performance Test Cases

A. Scalability Test

a. Assess RPL protocol performance as the number of nodes increases (e.g., 10, 20, 50, 100). This evaluates how well the system handles larger network sizes in terms of routing efficiency and stability.

B. Overhead Test

a. Compare energy consumption and control message overhead between standard RPL and the proposed trust-aware mechanism, highlighting trade-offs introduced by security enhancements.

C. Reliability Test

a. Measure critical metrics such as Packet Delivery Ratio (PDR) and throughput in both attack and defense scenarios, ensuring the proposed solution maintains acceptable service quality.

D. Accuracy Test

a. Evaluate the precision of the trust-based detection system by calculating detection accuracy along with false positive and false negative rates, ensuring malicious nodes are correctly identified without penalizing legitimate ones.

E. Resilience Test

a. Analyze the time required for the network to recover normal operation after attack mitigation, focusing on DODAG reconstruction, node reintegration, and restoration of key performance metrics.

5.0 Budget and Budget Justification

5.1 Hardware & Software Budget

Item	**Specification**	**Availability/Cost**
Development Laptop	≥16GB RAM, ≥512GB SSD, modern CPU	Already available – LKR 0
Contiki-NG OS	Open-source IoT simulation OS	Free
Cooja Simulator	Integrated with Contiki-NG	Free

Version Control (GitHub/Bitbucket)	Repository management	Free (student license)
Statistical Analysis Tools (Python, R)	Data processing and visualization	Free

Table 2 : Hardware & Software Budget

5.2 Research Resources

Item	**Requirement**	**Cost (LKR)**
University Library Access	IEEE Xplore, ACM DL, Springer (via institution)	0
Additional Digital Libraries	Institutional access	0
Cloud Computing Resources (if local capacity is insufficient)	50–100 hours of VM time for simulations	≈ LKR 30,000
Data Storage & Backup	Cloud-based archival (Google Drive/OneDrive)	Free (University quota)

Table 3: Research Resources

5.3 Contingency & Miscellaneous

Item	**Purpose**	**Cost (LKR)**
Printing/Documentation	Hard-copy reports, posters, diagrams	≈ LKR 6,000
Unexpected minor expenses	Backup tools, data transfer, etc.	≈ LKR 6,000

Table 3: Contingency & Miscellaneous

5.4Budget Justification

The proposed research leverages open-source tools (Contiki-NG, Cooja) and university-provided resources (IEEE Xplore, ACM Digital Library, cloud storage), which keeps costs very low.

- The only significant cost is cloud computing (≈LKR 30,000) to support large-scale simulations beyond laptop capacity, ensuring scalability and realistic disaster scenarios.
- A small contingency (≈LKR 15,000) is included for printing, documentation, and unforeseen requirements.
- No dedicated hardware purchases are necessary, as simulations are performed virtually, ensuring a lightweight, cost-effective research design suitable for resource-constrained IoT environments.

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